5th

Wednesday, September 2, 2026 9:49 AM

Nonhomogeneous dirichelet BCs

$$\begin{cases} -(au')' = f \quad \text{in } (0,1) \\ \text{at } u(0)=\alpha, \quad u(1)=\beta \end{cases}$$

Var. problem:

$$\int_0^1 au'v'\,dx - [au'v]_0^1 = \int_0^1 fv\,dx$$

To make boundary term vanish

$$V = \left\{ v : \int_0^1 [v^2 + (v')^2]\,dx < \alpha \; ; \; v(0)=\alpha \text{ and } v(1)=\beta \right\}$$

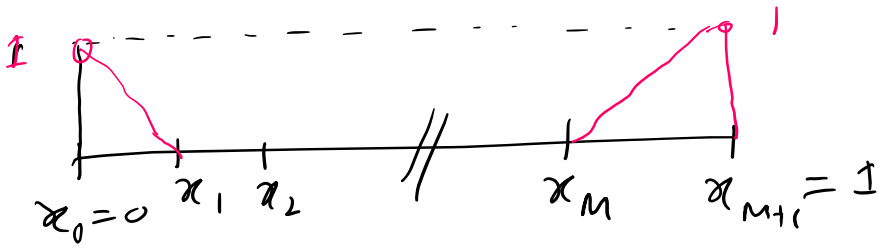
$$V_0 = \left\{ v : \int_0^1 [v^2+(v')^2]\,dx < \alpha \; ; \; v(0)=0=v(1) \right\}$$

Var. problem

Find $u \in V$ s.t

$$\int_0^1 au'v'\,dx = \int_0^1 fv\,dx \longrightarrow \forall \; v \in V_0$$

Nonhomogeneous Dirichelet BCs



use same basis : $\phi_j(x)$ $\quad j = 1, \dots, M$

$\Rightarrow$ introduce 2 "half-hat" functions :

$$\phi_0(x) = \frac{x_1 - x}{x_1 - x_0} \rightarrow x \in (x_0, x_1)$$

$$\phi_{M+1}(x) = \frac{x - x_M}{x_{M+1} - x_M} \rightarrow x \in [x_M, x_{M+1}]$$

Trial Function :

$$U(x) = \alpha \phi_0(x) + \beta \phi_{M+1}(x) + \sum_{j=1}^{M} \xi_j \cdot \phi_j(x)$$

Test functions :

$$V = \phi_i(x) \qquad i = 1, 2, \dots M$$

homogeneity means boundary is zero

$\alpha = \beta \neq 0$ — is not homogeneous

$\alpha = \beta = 0 \rightarrow$ homogeneous

Nonhomogeneous Dirichlet BCs

$$\int_0^1 a\, u' v'\, dx = \int_0^1 f v\, dx$$

$$\downarrow$$

$$\sum_{i=1}^{m} \xi_j \int_0^1 a \phi_i' \phi_j'\, dx + \alpha \int_0^1 a \phi_0' \phi_i'\, dx + \beta \int_0^1 a \phi_{m+1}' \phi_i'\, dx = \int_0^1 f \phi_i\, dx$$

$$\sum_{i=1}^{m} \xi_j \int_0^1 a \phi_i' \phi_j'\, dx = \int_0^1 f \phi_i\, dx + \delta_{1i} \frac{\alpha}{h^2} \int_{x_0}^{x_1} a(x)\, dx + \delta_{mi} \frac{\beta}{h_{m+1}^2} \int_{x_m}^{x_{m+1}} a(x)\, dx$$

Note: Dirichlet BCs are enforced strongly and modify the RHS → No error in BC from BCs

In FEM language: Dirichlet BCs for called essential BCs must be imposed in the definition of the

must be imposed in the definition of the function space V

Neumann BCs:

$$\begin{cases} -(au')' = f & \text{in } (0,1) \\ u(0) = 0 \\ a(1)\,u'(1) = g_1 \end{cases}$$

Here, $V = \left\{ v : \int (v^2 + (v')^2)\,dx < \infty, \; v(0) = 0 \right\}$

Int-by-parts:

$$\int_0^1 au'v'\,dx - \underbrace{a(1)u'(1)v(1)}_{g_1} + \underbrace{a(0)u'(0)v(0)}_{0,\ v(0)=0} = \int_0^1 fv\,dx$$

$$\Rightarrow \int_0^1 au'v'\,dx - g_1 v(1) = \int_0^1 fv\,dx \quad \forall\, v \in V$$

Q: is the solution to variational problem also a solution to 2-pt BVP?

also a solution to 2-pt BVP.

start with a var. pdh then reverse int-by parts

$$-\int_0^1 (au')' v\,dx + a(1)u'(1)v(1) = \int_0^1 fv\,dx + g_1 v(1)$$

This must hold for all $v \in V$, including all $v \in V$ with $v(1)=0$

$$\int_0^1 fv\,dx = -\int_0^1 (au')' v\,dx$$

$$\forall\ v \in V \cap v(1) = 0$$

$$\Rightarrow -(au')' = f \quad \text{in } (0,1)$$

So, $a(1)u'(1)v(1) = g_1 v(1)$

$$\Rightarrow a(1)u'(1) = g_1$$

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cG(1) FEM:

$V_1 = \{ v \in V : v|_I = \text{polynomial of degree}$

$V_h = \{ v \in V : v|_{[x_j, x_{j+1}]} = \text{polynomial of degree } 1 \}$

$V_h = \text{span}\{ \phi_1, \phi_2, \cdots \phi_m, \phi_{m+1} \}$

↑ new part

$\phi_{m+1}(x)$ is the half-hat function at end right end point.

Trial Function:

$$U(x) = \sum_{j=1}^{m+1} \xi_j \phi_j(x)$$

S.t.

$$\int_0^1 a U' v' dx = \int_0^1 f v dx + g_1 v(1) \quad \text{S.t } \forall v \in V_h$$

and $V = \phi_i \; \ell \quad i = 1, 2, \cdots m, m+1$

↑ new

$\phi_i(1) = 0 \quad \forall \; i = 1, 2, \cdots m$

$\phi_{m+1}(1) = 1$ ← only one that is not zero at $x = 1$

ψ_{m+1} ...

u

$x \geq 0$ at $x = 1$

at $x = 1$ what is the error?

$\downarrow$

$N_0 + x \geq 0$